Fitting Random Labels

Treat x_i as four email spam scores with $x_1 < x_2 < x_3 < x_4$. Replace their true labels by fair coin flips. One possible outcome is $\sigma = (+1, -1, +1, -1)$.

	x_1	x_2	x_3	x_4	Matches
Best constant rule	$+$	$+$	$+$	$+$	$2/4$
A best threshold rule	$-$	$-$	$+$	$+$	$2/4$
Unrestricted lookup rule	$+$	$-$	$+$	$-$	$4/4$

Here a threshold predicts $+1$ exactly when $x \geq a$.

$$\frac{1}{4}\sum_{i=1}^{4} \sigma_i h(x_i) = 2\left(\frac{\#\ \text{matches}}{4}\right) - 1.$$

How well can each class fit random labels *on average*, choosing a rule after each draw? The lookup class always achieves score 1.

Repeated and Distinct Inputs

Let a and b represent two patient profiles. Let $\mathcal{H}$ contain all four maps $\{a, b\} \to \{-1, +1\}$. Keep this class fixed and compare samples of size two.

	$S_1 = (a, a)$	$S_2 = (a, b)$
Available prediction vectors	$(+,+), (-,-)$	all four sign vectors
Best scores for $++, +-, -+, --$	$1, 0, 0, 1$	$1, 1, 1, 1$
Average over random signs	$\widehat{\mathfrak{R}}_{S_1} = 1/2$	$\widehat{\mathfrak{R}}_{S_2} = 1$

If a and b each have probability $1/2$, an i.i.d. sample repeats a point with probability $1/2$. The average over samples is

$$\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 = \frac{3}{4}.$$

The first average fixes the inputs. A second average accounts for which inputs the data distribution supplies.

Mixing Two Predictors

Fix one sample S and one sign vector σ. Suppose two real-valued functions have correlations

$$c_1 = \frac{1}{m}\sum_i \sigma_i g_1(z_i) = 0.2, \qquad c_2 = \frac{1}{m}\sum_i \sigma_i g_2(z_i) = 0.6.$$

For the mixture $f_\alpha = \alpha g_1 + (1-\alpha)g_2$, $0 \le \alpha \le 1$,

$$\frac{1}{m}\sum_i \sigma_i f_\alpha(z_i) = \alpha c_1 + (1-\alpha)c_2 = 0.6 - 0.4\alpha \in [0.2, 0.6].$$

The best mixture reaches 0.6, exactly the better endpoint score. We added infinitely many mixtures. Did we gain any ability to fit this noise vector? What happens when we average over all sign vectors?

This is a statement about real-valued convex mixtures; a majority-vote classifier is a different, nonlinear class.

Prediction Errors on Four Examples

On four labeled examples, take $y = (+1, +1, -1, -1)$ and compare two hypotheses:

i	y_i	$h_A(x_i)$	$\ell_{0\text{-}1}(h_A(x_i), y_i)$	$h_B(x_i)$	$\ell_{0\text{-}1}(h_B(x_i), y_i)$
1	$+1$	$+1$	0	$+1$	0
2	$+1$	$+1$	0	-1	1
3	-1	-1	0	-1	0
4	-1	$+1$	1	-1	0

Thus the two predictors produce loss vectors $(0, 0, 0, 1)$ and $(0, 1, 0, 0)$; each has training error $1/4$.

Generalization asks whether these average errors remain small on new data. Which functions should our complexity analysis evaluate to control errors?

Swapping Two Samples

Take two independent samples $S, S' \sim \mathcal{D}^2$. For a fixed g, suppose the losses are

i	$g(z_i)$	$g(z_i')$	$d_i = g(z_i') - g(z_i)$
1	0.1	0.7	$+0.6$
2	0.4	0.2	-0.2

$$\widehat{\mathbb{E}}_{S'} g - \widehat{\mathbb{E}}_S g = \tfrac{1}{2}(d_1 + d_2) = 0.20.$$

Swapping either member of a pair multiplies its difference by a sign:

$$\tfrac{1}{2}(\sigma_1 d_1 + \sigma_2 d_2), \qquad \sigma_i \in \{-1, +1\}.$$

For example, $(\sigma_1, \sigma_2) = (+1, -1)$ gives 0.40.

Both samples come from the same distribution. Randomly swapping each pair preserves their joint distribution and introduces random signs. Can this turn a comparison of sample averages into a noise-fitting problem?

The second sample is a proof device. The numerical gap can change under a swap, while its distribution remains unchanged.

Many Cutoffs, Five Predictions

A sensor raises an alarm when $x \geq a$. Four observed readings are $1, 2, 3, 4$, and the cutoff a may be any real number.

Cutoff	$x = 1$	$x = 2$	$x = 3$	$x = 4$
$a \leq 1$	1	1	1	1
$1 < a \leq 2$	0	1	1	1
$2 < a \leq 3$	0	0	1	1
$3 < a \leq 4$	0	0	0	1
$a > 4$	0	0	0	0

Cutoffs $a = 2.1$ and $a = 2.9$ are different rules but agree on these data.

On this sample, infinitely many rules produce only five prediction vectors. What should we count when we measure the class on finite data?

101 Vectors, Many Sign Draws

Consider increasing thresholds on 100 distinct ordered inputs. Encode their predictions as -1 and $+1$.

Quantity	Value
Possible real cutoffs	infinitely many
Distinct prediction vectors	101
Euclidean norm of each prediction vector	$\sqrt{100} = 10$
Possible random sign vectors	$2^{100} \approx 1.27 \times 10^{30}$

Exact averaging over all sign vectors is impractical even in this example.

Can the number of prediction vectors and their Euclidean norms bound their best average correlation with random signs?

An Interval Labeling Game

A classifier predicts 1 inside one interval and 0 outside. Think of x as a biomarker or dosage level and the interval as a target range. Imagine that someone chooses the labels *after* we place the points.

Inputs	Labels to fit	Can one interval fit them?
$x_1 < x_2$	$00, 01, 10, 11$	Yes, all four choices
$x_1 < x_2 < x_3$	010	Yes, keep only the middle point
$x_1 < x_2 < x_3$	101	No, an interval cannot skip the middle

The labeling 010 is feasible. Our challenge is to fit *every* labeling of the same points.

What is the largest number of points for which our class can win against every possible choice of labels?

Counting Interval Patterns

Intervals cannot fit 101 on three ordered inputs. Their positive labels must form one contiguous block, giving $1 + m(m+1)/2$ patterns.

Number of inputs m	Interval patterns	All binary patterns 2^m
3	7	8
5	16	32
10	56	1,024
20	211	1,048,576

Missing one pattern on every triple accompanies a large restriction on longer label sequences.

If a different class also has VC dimension 2, must it obey a similar bound even without the geometry of intervals?

Extending a Prediction Pattern

Increasing thresholds on $x_1 < x_2 < x_3$ produce

$$G = \{000, 001, 011, 111\}.$$

Remove the last coordinate and group the remaining prefixes.

Prefix on (x_1, x_2)	Allowed last labels	Full patterns
00	0 or 1	000, 001
01	1	011
11	1	111

There are three prefixes, and one prefix has an extra extension:

$$4 = 3 + 1.$$

Count each prefix once, then add the prefixes that allow both labels at the last point. This is the split used in Sauer's proof.

A Data Budget for Interval Rules

Suppose we want every interval rule's population error to be at most its training error plus 0.10, with confidence 95%. For $d = 2$ and $\delta = 0.05$, the bound already derived gives the penalty

$$B(m) = \sqrt{\frac{4\log(em/2)}{m}} + \sqrt{\frac{\log 20}{2m}}.$$

Training examples m	Penalty $B(m)$	At most 0.10?
1,000	0.2086	No
5,000	0.1013	No
6,000	0.0933	Yes

Instead of inserting m into a bound, solve for an m that makes the penalty small enough. This gives a sufficient data budget.

Unseen Labels

Suppose $\mathcal{H}$ shatters four inputs $\{a, b, c, d\}$. Choose a target uniformly from the 16 possible labelings on these inputs.

Input	a	b	c	d
Observed in training?	Yes	Yes	No	No
Observed label	$+$	$-$	?	?

The four possibilities for the unseen pair are

$$(y_c, y_d) \in \{(+,+),(+,-),(-,+),(-,-)\}.$$

All remain equally likely. Whatever the learner predicts, its expected error on each unseen input is $1/2$ under this random target.

Even a perfect fit to observed labels cannot determine unseen labels. Assigning probability mass to rarely seen inputs creates a lower bound that applies to every learning algorithm.

A Nearly Fair Label Coin

At one medical test setting, labels are noisy. The learner must distinguish

$$\mathbb{P}(y = +1 \mid x) = 0.55 \qquad \text{from} \qquad \mathbb{P}(y = +1 \mid x) = 0.45.$$

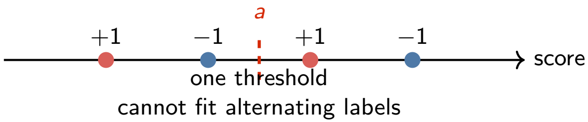
	$p = 0.55$	$p = 0.45$
Best constant prediction	$+1$	-1
Best error	0.45	0.45
Error with the wrong prediction	0.55	0.55
Excess error	0.10	0.10

With n labels, the empirical positive fraction has standard deviation $\sqrt{p(1-p)/n} \approx 0.5/\sqrt{n}$.

For biases $1/2 \pm \gamma$, halving the bias needs about four times as many labels to keep the same signal-to-noise ratio. The lower-bound construction repeats this difficulty across d shattered inputs.

Visual: Spam Score and Random Labels

Treat the horizontal coordinate as a spam score. The observed labels alternate, so one increasing cutoff cannot separate them all.



Lookup rule

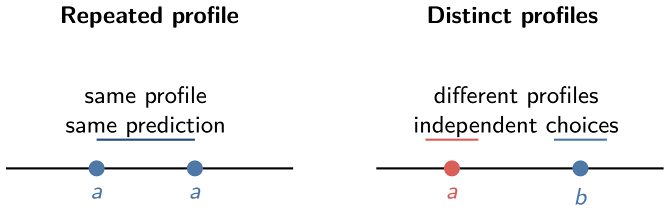
stores one decision per email

fits all four labels

Question: which model class can correlate with random labels after seeing them?

Visual: Repeated and Distinct Patients

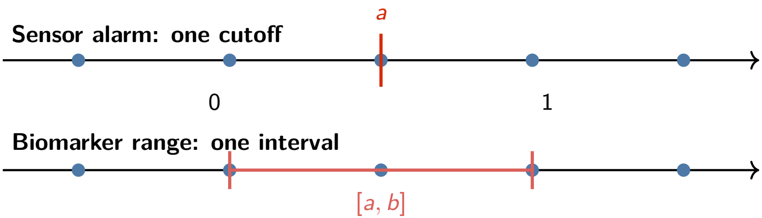
The same risk model can have different flexibility on different samples.



The first sample forces repeated evaluations to agree. The second sample lets the class choose two predictions separately.

Visual: Threshold and Interval Rules

Two common rules over a one-dimensional measurement have different shapes.

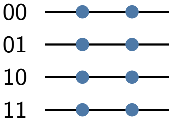


Infinitely many parameter values can produce only finitely many behaviors on the observed points.

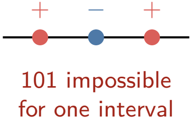
Visual: Shattering as a Labeling Game

Place the points first. Then ask whether the class can answer every labeling challenge.

Two points: all four challenges



Three points: one challenge fails

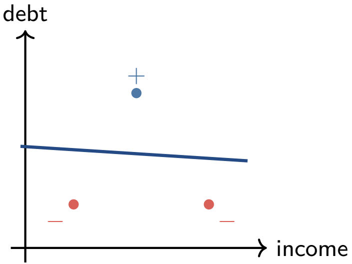


Every labeling can be realized by a suitable interval.

The largest number of points on which every labeling is possible is the VC dimension.

Visual: Linear Loan Boundary

Let the axes represent income and debt. A linear approval rule is one side of a line.



Three non-collinear customer profiles can realize every binary approval pattern by moving the line. Four profiles create a geometric obstruction, such as XOR.

This is the geometric intuition behind VC dimension 3 for affine separators in $\mathbb{R}^2$.